

Integrated modeling of helium-vacancy clustering in Eurofer97 steel upon $\text{He}^+/\text{Fe}^{3+}$ dual-beam irradiation

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ABSTRACT

By developing an integrated rate theory model, we studied helium-vacancy (He-V) clustering in ferritic/martensitic Eurofer97 steels under $\text{He}^+/\text{Fe}^{3+}$ dual-beam irradiation. Different from the known single peak concentration profile of total atomic He, the concentration of 'solute' He atom (interstitial He) increases and decreases twice during irradiation, showing two peaks as a function of irradiation time (dose). The second increase in concentration was caused by the operation of the kick-out mechanism aided by irradiation-induced self-interstitials. We show that the kicked-out interstitial He enhances the growth of He-V clusters and leads to the shift of cluster pressurization states. He-V clustering direction is also governed by the kicked-out interstitial He.

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1. Introduction

Helium bubbles or helium-vacancy (He-V) clusters produced in metals and alloys are typical radiation defects that can lead to swelling, hardening, and embrittlement of irradiated materials [1–4]. The evolution of He bubbles under continuous He implantation or neutron irradiation in metals can be generally characterized by three periods (incubation, nucleation and growth) [5], as schematically shown in Fig. 1. It can be safely assumed that the implanted or neutron transmuted He atoms are initially in the interstitial sites. Because of the high binding energy between He atoms and vacancies (2.3 eV in body-centered cubic (bcc) iron [6]), interstitial He can be easily trapped by vacancies and thus becomes substitutional He. In the bubble incubation period, the total atomic He (interstitial and substitutional He) accumulates to a high concentration, which contributes to the rapid bubble nucleations in the bubble nucleation period.

Because of the low migration energy (0.06 eV in bcc iron [6]), interstitial He plays an important role in bubble nucleation and growth periods. Intrinsically, substitutional He is immobile. However, a substitutional He atom can enter interstitial sites, through two possible mechanisms: the Frank-Turnbull [7,8] and kick-out mechanisms [8,9]. The first one occurs when a substitutional He (HeV) moves to an interstitial site, leaving a vacancy behind ($\text{HeV} \rightarrow \text{He}_{\text{int}} + \text{V}$). This mechanism is governed by the dissociation

of HeV and needs a relatively high temperature to activate. The latter operates when a self-interstitial (SIA) displaces a substitutional He into an interstitial site ($\text{HeV} + \text{SIA} \rightarrow \text{He}_{\text{int}}$). From an energetic point of view, the activation of the kick-out mechanism is expected to be spontaneous when a SIA approaches HeV [9]. During the initial phase of irradiation, the anticipated accumulation of SIAs and HeV implies the increasing occurrence of kick-out reactions. Due to the faster migration of SIAs to existing sinks in the matrix compared with vacancies, the level-off concentration of vacancies would be higher than SIAs [3]. The high concentration of vacancies may cancel the effect of the kick-out reactions by re-trapping the kicked-out He atoms, before they migrate to the cluster embryos. It implies that the competition exists between the kick-out and re-trapping reactions. The competition is expected to be strong in the bubble nucleation and growth periods, and is sensitive to the concentrations of SIAs and vacancies. Though the general trends of main quantities characterizing He bubble evolution during irradiation are qualitatively known (Fig. 1), how does the kick-out mechanism affect the evolution of the two forms of He, and further the kinetics of He-V clustering is not quantitatively understood.

To clarify the above issues, the dynamical evolution of irradiation-induced single defects and defect clusters needs to be addressed. The cluster dynamics modeling based on the rate theory has been widely used in this aspect, and proved to be a powerful method [8,10–13]. In this work, we established a rate theory framework to study the kinetics of He-V clustering in reduced activation ferritic/martensitic Eurofer97 steel, which is a promising structure material candidate for future fusion reactors [14–17]. For

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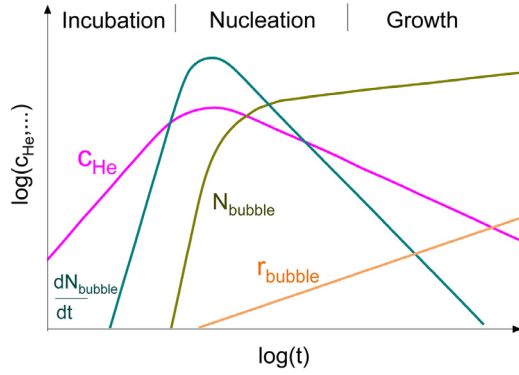


Fig. 1. Schematic representation of the time-dependence of the main quantities characterizing He bubble evolution during irradiation, following Ref. [5]. C_{He} is the concentration of atomic He (interstitial and substantial He); N_{bubble} is the He bubble density; dN_{bubble}/dt is the bubble nucleation rate; r is the average size of bubbles.



Fig. 2. Schematic drawing of the coupled evolution between CG2D and CG1D. β describes the dynamical He/V ratio of clusters in CG1D. V_c is the critical vacancy number of clusters in CG2D.

the energetics of He-V clusters, a variable-gap model (VGM) [18] is applied to determine the binding energies of large He-V clusters (Supplementary note 1.1). For small clusters, $He_{m \leq 5}V_{n \leq 5}$, we adopt the values optimized based on the thermal helium desorption experiments [12,19]. Similar to the model of Brimbal et al. [11], a coupled evolution scheme between a 2-dimensional He-V cluster group (CG2D) and 1-dimensional He-V cluster group (CG1D) was developed, as schematically shown in Fig. 2. The CG2D has a domain size of 20 (number of vacancy) $\times$ 50 (number of He atom), used to describe the He-V nucleation and initial evolution phase. In CG1D, the number of vacancies within He-V clusters is over 20. The CG1D is designed to describe the cluster growth phase. This work simulates the He-V clustering in Eurofer97 under He⁺/Fe³⁺ dual-beam irradiation [15], with an equivalent dpa (displacements per atom) rate of 6.94×10^{-4} dpa/s, and He generation rate of 1.20×10^{-2} appm/s at irradiation temperature of 673 K. The production ratio of He atom to dpa simulates the impact of neutron irradiation on Eurofer97 expected in future fusion reactors.

2. Rate theory framework

The rate theory model is built based on the principle that the evolution of clustering objectives is described by their variations in population and configuration with time. The evolution proceeds through the ‘chemical kinetics’ in a homogeneous medium, where the probability of a cluster with configuration u to become a cluster of $u+x$ or $u-x$ depends on its rate of absorbing or emitting corresponding mobile specimen x . The general idea and approach have been described in detail and validated elsewhere [3,8,10,12,13,20].

2.1. Mobile defects

The point defects (He, V and SIA), di-SIA (I_2) and tri-SIA (I_3) are considered to be mobile, and only point defects can be emitted from defect clusters. The mobility of I_2 and I_3 is taken into consideration because they have similar ground-state configurations and comparable migration abilities compared with SIA in bcc iron [21]. Larger interstitial clusters/loops containing over 3 SIAs are considered to be immobile due to the trapping by impurities and He atoms [22–24]. Similarly, the migration of pure vacancy clusters and He-V clusters is ignored due to the possible trapping by carbon as well as Cr atoms in Eurofer97 steels [25,26]. These assumptions will be discussed later separately.

For the diffusion of substitutional He, besides the aforementioned Frank-Turnbull [7,8] and kick-out mechanisms [8,9], substitutional He can also diffuse in iron by the vacancy mechanism via the migration of HeV₂ complexes [6]. However, the migration energy (1.1 eV) of HeV₂ complexes is much higher than the one (0.06 eV) of He_{int}. Moreover, with a V-HeV binding energy of 0.78 eV, HeV₂ may dissociate rapidly at elevated temperatures. Recently, Brimbal et al. [11] reported the small effect of HeV₂ mobility on cavity density and size at temperatures over 400 °C. Therefore, the mobility of HeV₂ is not considered in this model. We notice that the same assumption has been made in the work of Ortiz et al. [8].

2.2. Binding energies of He-V clusters

For small He-V clusters, $He_{m \leq 5}V_{n \leq 5}$ play vital roles in cluster nucleation and further evolution. In the work of Xu et al. [12], the binding energies of these clusters were optimized based on thermal helium desorption spectrometry (THDS). Importantly, the validation of these energetics has been examined in a wide temperature regime. In this work, we adopt the values of optimized binding energies ($He_{m \leq 5}V_{n \leq 5}$) given in Ref. [12].

The binding energies of larger He-V clusters are calculated based on Jourdan’s variable gap model (VGM) [18], where the free energy of clusters was parametrized based on molecular dynamics (MD) calculations performed in bcc iron. The strong He-iron repulsion induced interfacial gap between He-V cluster and surrounding iron matrix was considered in the determination of formation free energy $E_f(m, n)$ of cluster He_mV_n . The binding energy of the cluster He_mV_n to a vacancy, He atom and SIA is calculated respectively as

$$E_b^V(m, n) = E_f(0, 1) + E_f(m, n-1) - E_f(m, n), \quad (1)$$

$$E_b^{He}(m, n) = E_f(1, 0) + E_f(m-1, n) - E_f(m, n), \quad (2)$$

$$E_b^{SIA}(m, n) = E_f^{SIA} + E_f(m, n+1) - E_f(m, n), \quad (3)$$

where $E_f(0, 1) = E_f^V$, $E_f(1, 0) = E_f^{He}$ and E_f^{SIA} are the formation energy of a vacancy, interstitial He atom and SIA in bcc iron, respectively. Using VGM, the calculated binding energies and dissociation energies of defects (He, SIA and vacancy) showed good consistency with previous MD [27] and density function theory (DFT) studies [6]. For the experimental validation of these binding energies, the solid-fluid transition of He bubbles was evaluated based on VGM, the deduced melting curve also showed good agreement with available experimental data [28,29]. Interested readers are encouraged to refer to Ref. [18] for more details.

2.3. Governing equations of point defects and defect clusters

The model contains a series of coupled ordinary differential equations (ODEs) that define the evolution of single defects and

defect clusters. The density of He-V clusters He_mV_n is denoted as $C(m, n)$; For self-interstitial clusters/loops $\text{SIAs}(i)$ consisting of i SIAs, the density is denoted as $\text{CSIAS}(i)$. The governing equation for vacancy and SIA, i.e., $\frac{dC(0,1)}{dt}$ and $\frac{d\text{CSIAS}(1)}{dt}$ respectively, can be structured as

$$\frac{dC(0,1)}{dt} = G_{\text{irra}}^V - L_{\text{sink}}^V + [E_{\text{He-V}}^V - A_{\text{He-V}}^V + A_{(0,3)}^{I_2} + A_{(0,4)}^{I_3}] + [E_{\text{SIAs}}^V - A_{\text{SIAs}}^V], \quad (4)$$

$$\frac{d\text{CSIAS}(1)}{dt} = G_{\text{irra}}^{\text{SIA}} - L_{\text{sink}}^{\text{SIA}} + [E_{\text{He-V}}^{\text{SIA}} - A_{\text{He-V}}^{\text{SIA}} + A_{(0,1)}^{I_2} + A_{(0,2)}^{I_3}] + [E_{\text{SIAs}}^{\text{SIA}} - A_{\text{SIAs}}^{\text{SIA}}], \quad (5)$$

where G_{irra}^x ($x = V$ and SIA) is generation rate of defect x from irradiation; L_{sink}^x is loss rate of defect x to sinks; $E_{\text{He-V}}^x$ and $A_{\text{He-V}}^x$ are emission and absorption rates of defect x by He-V clusters, respectively. E_{SIAs}^x and A_{SIAs}^x are emission and absorption rates of defect x by interstitial clusters/loops, respectively. $A_{(0,n)}^{I_2}$ and $A_{(0,n)}^{I_3}$ describe the production rates of vacancy (and SIA) resulting from the absorption of I_2 and I_3 by pure vacancy clusters, respectively. The governing equations for the mobile interstitial He, I_2 and I_3 can be expressed similarly. For interstitial He, however, the term of He loss to sinks is ignored. This assumption was made following the treatment by Xu et al. [12], who considered that He migrating to sinks cannot lose its identity and the uncertainty exists in the strength of the interaction between He atoms and dislocations according to atomistic calculations [30].

Based on the above formalism, the governing equations for the evolution of immobile He-V clusters and self-interstitial clusters/loops (considered as prismatic dislocation loops here) can be written as

$$\begin{aligned} \frac{dC(m,n)}{dt} = & G_{\text{irra}}^{(m,n)} - \sum_{x=V, \text{SIA}, \text{He}, I_2, I_3} k_{(m,n)}^x C(m,n) C(x) \\ & - \sum_{x=V, \text{SIA}, \text{He}} g_{(m,n)}^x C(m,n) \\ & + \sum_{x=V, \text{SIA}, \text{He}, I_2, I_3} k_{(m', n')}^x C(m', n') C(x) \\ & + \sum_{x=V, \text{SIA}, \text{He}} g_{(m', n')}^x C(m', n'), \end{aligned} \quad (6)$$

$$\begin{aligned} \frac{d\text{CSIAS}(i)}{dt} = & G_{\text{irra}}^{\text{SIA}(i)} - \sum_{x=V, \text{SIA}, I_2, I_3} k_{\text{SIAs}(i)}^x \text{CSIAS}(i) C(x) \\ & - \sum_{x=V, \text{SIA}} g_{\text{SIAs}(i)}^x \text{CSIAS}(i) \\ & + \sum_{x=V, \text{SIA}, I_2, I_3} k_{\text{SIAs}(i)}^x \text{CSIAS}(i) C(x) \\ & + g_{\text{SIAs}(i+1)}^{\text{SIA}} \text{CSIAS}(i+1) + g_{\text{SIAs}(i-1)}^V \text{CSIAS}(i-1), \end{aligned} \quad (7)$$

with

$$k_{(m,n)}^x = 4\pi [D_x + D_{(m,n)}] [r_x + r_{(m,n)}], \quad (x = \text{He}, V \text{ and SIA}) \quad (8)$$

$$g_{(m,n)}^x = \frac{1}{\Omega} k_{(m,n)}^x \exp[E_b^x(m,n)/k_B T], \quad (x = \text{He}, V \text{ and SIA}) \quad (9)$$

$$k_{\text{SIAs}(i)}^x = 2\pi [D_x + D_{\text{SIAs}(i)}] [2r_x + r_{\text{SIAs}(i)}] Z_{\text{SIAs}(i)}^x, \quad x = V \text{ and SIA} \quad (10)$$

$$g_{\text{SIAs}(i)}^x = \frac{1}{\Omega} k_{\text{SIAs}(i)}^x \exp[E_{\text{SIAs}(i)}^x/k_B T], \quad (x = V \text{ and SIA}) \quad (11)$$

where $k_{(m,n)}^x$ and $g_{(m,n)}^x$ are the reaction rate of cluster He_mV_n absorbing and emitting defect x , respectively. The second and third terms on the right-hand-side of Eq. (6) describe the depletion of He_mV_n due to their reactions with mobile defects. The last two terms describe the generation of He_mV_n from the reactions between their configuration-nearby clusters (He_mV_n or He_mV_n) and mobile defects. Similar explanations apply to the governing equations of dislocation loops. $G_{\text{irra}}^{(m,n)}$ and $G_{\text{irra}}^{\text{SIAs}(i)}$ are respectively generation rates of small vacancy clusters ($m=0, n \leq 4$) and interstitial clusters ($i \leq 4$) directly from the intra-cascade clustering [20,31,32].

$$G_{\text{irra}}^{(0,n)} = \frac{\eta G_{\text{dpa}} f_V(n)}{n}, \quad n \leq 4, \quad (12)$$

$$G_{\text{irra}}^{\text{SIAs}(i)} = \frac{\eta G_{\text{dpa}} f_{\text{SIAs}(i)}}{i}, \quad i \leq 4, \quad (13)$$

where η is damage efficiency; G_{dpa} is damage rate; $f_V(n)$ and $f_{\text{SIAs}(i)}$ denote the formation fractions of $V(n)$ and $\text{SIA}(i)$ after each irradiation cascade. Here, $f_V(4)$ and $f_{\text{SIAs}(4)}$ are set to include the contributions from all larger clusters with components (vacancy or SIA) more than 4.

The absorption and emission rate defined in Eqs. (8-11) adopt the formalism in Ref. [33]. D and r generally denote the diffusivities and radii of reactants, respectively. $E_b^x(m,n)$ is the binding energy of cluster He_mV_n to point defect x ; $E_{\text{SIAs}(i)}^{\text{SIA}}$ is the binding energy of $\text{SIAs}(i)$ to a SIA, and is estimated using the extrapolation law [20]

$$E_{\text{SIAs}(i)}^{\text{SIA}} = E_f^{\text{SIA}} + [E_{\text{SIAs}(2)}^{\text{SIA}} - E_f^{\text{SIA}}] \left[i^{\frac{2}{3}} - (i-1)^{\frac{2}{3}} \right] / \left(2^{\frac{2}{3}} - 1 \right). \quad (14)$$

For the binding energy of $\text{SIAs}(i)$ to a vacancy, $E_{\text{SIAs}(i)}^V$ is calculated as [23]

$$E_{\text{SIAs}(i)}^V = E_f^V + E_f^{\text{SIA}} - E_{\text{SIAs}(i+1)}^{\text{SIA}}. \quad (15)$$

$Z_{\text{SIAs}(i)}^x$ in Eq. (10) is the bias factor of $\text{SIAs}(i)$ absorbing point defects (vacancy and SIA), and is estimated using the analytical formula established by Jourdan [34] as

$$Z_{\text{SIAs}(i)}^x = \frac{2\pi \left\{ 1 + \frac{1}{2} \left(\frac{r_0^{\text{eff}}}{r_{\text{SIAs}(i)}^{\text{eff}}} \right) \left[\ln \left(\frac{8r_{\text{SIAs}(i)}^{\text{eff}}}{r_0^{\text{eff}}} \right) - \frac{3}{2} \right] \right\}}{\ln \left(\frac{8r_{\text{SIAs}(i)}^{\text{eff}}}{r_0^{\text{eff}}} \right) + \frac{1}{4} \left(\frac{r_0^{\text{eff}}}{r_{\text{SIAs}(i)}^{\text{eff}}} \right)^2 \left\{ \left[\ln \left(\frac{8r_{\text{SIAs}(i)}^{\text{eff}}}{r_0^{\text{eff}}} \right) \right]^2 - \ln \left(\frac{8r_{\text{SIAs}(i)}^{\text{eff}}}{r_0^{\text{eff}}} \right) - \frac{5}{4} \right\}}}, \quad (16)$$

$$r_0^{\text{eff}} = \alpha_1 L + 2 \exp(-\alpha_2 L^{\alpha_3}) - \frac{\beta_1 L^{\beta_2}}{\frac{\beta_3}{L^{\beta_4}} + \exp \left[\frac{r_{\text{SIAs}(i)}^{\beta_5}}{L^{\beta_6}} \right]}, \quad (17)$$

$$L = \frac{1+\nu}{3\pi(1-\nu)} \frac{\mu b}{k_B T} \Delta\Omega_x, \quad (18)$$

$$r_{\text{SIAs}(i)} = \left(\frac{i\Omega}{\pi b} \right)^{\frac{1}{2}}, \quad (19)$$

where ν , μ and b are Poisson's ratio, shear modulus and magnitude of the Burgers vector, respectively; $\Delta\Omega_x$ is relaxation volume of the point defect x ; Values of α_{1-3} and β_{1-6} are given in Ref. [34].

Mobile defects loss to existing sinks, L_{sink}^x , includes the loss to grain boundaries (GBs) and dislocation lines with sink strengths ω_{gb}^x and $Z_{\text{dl}}^x \cdot \rho_{\text{dl}}$, respectively.

$$L_{\text{sink}}^x = D_x (C_x - C_{\text{eq}}^x) (\omega_{\text{gb}}^x + Z_{\text{dl}}^x \cdot \rho_{\text{dl}}), \quad (20)$$

where C_{eq}^x is the thermal equilibrium concentration of point defect x ; Z_{dl}^x is bias factor of dislocation lines absorbing point defect x ; ρ_{dl} is the density of dislocation lines.

For defects loss to GBs, the expression of GB sink strength $\omega_{gb}^x = 15 / R_{gb}^2$, depending solely on grain size R_{gb} , was adopted previously [15]. A recent molecular statics simulation [35], however, unveils that GBs in bcc iron are biased sinks, favorite of trapping SIAs over vacancies. Furthermore, it was reported that sink strength of low-angle GBs, relying on misorientation angles, is stronger than that of high-angle GBs [36,37]. Applying the misorientation-angle-dependent GB sink strength, the intergranular irradiation-induced segregation was successfully predicted in Ref. [38]. Hence, the expression of sink strength for GBs should account for the microstructural nature of the material under investigation. A distinctive feature associated with Eurofer97 steel is the existence of a high density of lath structures, and the intralath misorientations are generally in the low-angle regime [39,40]. Here, we adopt a sophisticated expression, which describes the sink strength of low-angle GBs for point defects by taking into account GB microstructures [41]

$$\omega_{gb}^x = \frac{2\pi\theta}{\ln \frac{2}{\pi\theta} - h_x} \frac{1}{b}, \quad (21)$$

$$h_x = \ln \left(\frac{\beta_x b}{r_c^x} \right) + \frac{K_0 \left(\frac{\beta_x b}{2r_c^x} \right)}{I_0 \left(\frac{\beta_x b}{2r_c^x} \right)} + 0.5772, \quad (22)$$

$$\beta_x = \frac{\mu \Delta \Omega_x (1 + \nu)}{3\pi k_B T (1 - \nu)}, \quad (23)$$

where θ is GBs misorientation angle; K_0 and I_0 are the modified Bessel functions of the first and second kind, respectively. r_c^x is dislocation core radius for point defect x .

For the bias factor of dislocations absorbing point defects, constant parameters of $Z_{dl}^V = 1$ and $Z_{dl}^{SIA} = 1.2$ were commonly adopted [20,42], for instance in the model of Mamivand et al., though they considered the evolution of network dislocations [20]. In this work, we assume that the density of dislocation lines keeps unchanged during irradiation. It means that the variation of network dislocation density due to the possible growth and unfaulting of Frank loops, activation of Bardeen-Herring sources and irradiation-enhanced climb of dislocations was ignored for Eurofer97 steels. With the constant dislocation density ρ_{dl} , Z_{dl}^x was estimated using the analytical formula of Rauh & Simon [34,43].

$$Z_{dl}^x = 2\pi \sum_{n=0}^{\infty} (2 - \delta_{n,0}) (-1)^n \frac{I_n \left(\frac{L}{2r_c^x} \right) I_n \left(\frac{L}{2R} \right)}{I_n \left(\frac{L}{2r_c^x} \right) K_n \left(\frac{L}{2R} \right) - I_n \left(\frac{L}{2R} \right) K_n \left(\frac{L}{2r_c^x} \right)}, \quad (24)$$

$$\delta_{n,0} = \begin{cases} 1, & n = 0 \\ 0, & n > 0 \end{cases} \quad (25)$$

$$R = 1 / \sqrt{\pi \rho_{dl}}. \quad (26)$$

The above formalisms are used to determine the efficiency of sink structures (GBs, dislocations and dislocation loops) capturing typical point defects (vacancy and SIA). For the mobile self-interstitial clusters (I_2 and I_3 in this study), however, the analytical formulas are invalid. Since I_2 and I_3 have a similar ground-state configuration (aligned $\langle 110 \rangle$ dumbbells), here we simply assume $\omega_{gb}^{I_2} = \omega_{gb}^{I_3} = \omega_{gb}^{SIA}$, $Z_{SIA(i)}^{I_2} = Z_{SIA(i)}^{I_3} = Z_{SIA(i)}^{SIA}$ and $Z_{dl}^{I_2} = Z_{dl}^{I_3} = Z_{dl}^{SIA}$. Dedicated atomic simulation for the interaction between self-interstitial clusters and sink structures is required to further precisely determine relevant parameters.

2.4. Coupling scheme of two He-V cluster groups

The present rate theory framework applies a coupling scheme to connect the He-V clustering between two size-scale regimes, as schematically shown in Fig. 2. In CG2D, He-V clusters $He_m V_n$ have both the vacancy and He identities (i.e., m and n , respectively). The vacancy identity (number) n of clusters in CG2D are smaller than V_c (set as 20 in this work). The clusters with vacancy numbers over V_c would enter CG1D and lose their He identities. For the cluster containing vacancy number n_V in CG1D, the corresponding density is denoted as $C(n_V)$. The differential equation for general clusters ($n_V > V_c + 1$) in CG1D can be written as

$$\begin{aligned} \frac{dC(n_V)}{dt} = & k_{(n_V-1)}^V C(n_V - 1) C(0, 1) - k_{(n_V)}^V C(n_V) C(0, 1) \\ & - g_{(n_V)}^V C(n_V) + g_{(n_V+1)}^V C(n_V + 1) + k_{(n_V+1)}^{SIA} C(n_V + 1) \\ & CSIA_s(1) - k_{(n_V)}^{SIA} C(n_V) CSIA_s(1) - g_{(n_V)}^{SIA} C(n_V) \\ & + g_{(n_V+1)}^{SIA} C(n_V + 1) - k_{(n_V)}^{I_2} C(n_V) C_{I_2} + k_{(n_V+2)}^{I_2} C(n_V + 2) \\ & C_{I_2} - k_{(n_V)}^{I_3} C(n_V) C_{I_3} + k_{(n_V+3)}^{I_3} C(n_V + 3) C_{I_3}, \end{aligned} \quad (27)$$

where $C_{I_2} = CSIA_s(2)$, $C_{I_3} = CSIA_s(3)$ are the concentration of I_2 and I_3 , respectively. For critical clusters containing vacancy number $V_c + 1$, they can form when the clusters with vacancy number of V_c in CG2D absorb vacancy or emit SIA, as shown in Fig. 2. The differential equation, $\frac{dC(V_c+1)}{dt}$, needs to add the corresponding reaction terms: $k_{(m, V_c)}^V C(m, V_c) C(0, 1)$ and $g_{(m, V_c)}^{SIA} C(m, V_c)$. The emission rate ($g_{(n_V)}^x$, $x = He, V$ and IA) of clusters in CG1D will depend on the vacancy number (n_V) of the cluster and the cluster He/V ratio, which is estimated below.

We notice that in the model of Brimbal et al. [11], a similar coupling scheme was applied. The number of He atoms in clusters containing n_V vacancies in the 1D regime was averaged through the total number of He atoms contained in all clusters with n_V vacancies. In this model, the number of He atoms of each cluster in CG1D is estimated based on an assumption that cluster He/V ratio is identical for all clusters in CG1D, but dynamically updated by the ratio of total He to vacancy numbers in CG1D. Therefore, the virtual He identity of a cluster in CG1D can be estimated by multiplying the dynamical He/V ratio with the cluster vacancy number. The differential equation for the total He contents in CG1D can be written as

$$\begin{aligned} \frac{dC_{CG1D}^{He}}{dt} = & \sum_{n_V} k_{(n_V)}^{He} C(n_V) C(He_{int}) - \sum_{n_V} g_{(n_V)}^{He} C(n_V) \\ & + \sum_m m \cdot k_{(m, V_c)}^V C(m, V_c) C(0, 1) \\ & - R_{He/V}(V_c) \cdot (V_c + 1) \cdot g_{(V_c+1)}^V C(V_c + 1) \\ & + \sum_m m \cdot g_{(m, V_c)}^{SIA} C(m, V_c) \\ & - R_{He/V}(V_c) \cdot (V_c + 1) k_{(V_c+1)}^{SIA} C(V_c + 1) \\ & - \sum_{n_V=V_c+1, V_c+2} R_{He/V}(n_V - 2) \cdot n_V \cdot k_{(n_V)}^{I_2} C(n_V) \\ & - \sum_{n_V=V_c+1, V_c+2, V_c+3} R_{He/V}(n_V - 3) \cdot n_V \cdot k_{(n_V)}^{I_3} C(n_V), \end{aligned} \quad (28)$$

where $R_{He/V}(n)$ is the ratio of He to vacancy number in the cluster subgroup CSG(n), wherein the clusters contain the identical vacancy number n in CG2D. In this way, the He content in CG1D can be dynamically monitored. The vacancy content in CG1D, C_{CG1D}^V , can be directly calculated based on the cluster density in CG1D as

$C_{CG1D}^V = \sum_{n_V} n_V \cdot C(n_V)$. Therefore, the dynamical He/V ratio in CG1D can be estimated as $\beta = \frac{C_{CG1D}^{He}}{C_{CG1D}^V}$ during evolution.

He-V clusters containing vacancy numbers of V_c , $V_c - 1$ and $V_c - 2$ are critical clusters in CG2D. The differential equation for these clusters (based on the Eq. (6)) also needs to add the reaction terms, which account for the clusters in CG1D evolving back to CG2D via absorbing SIA or/and I_2 and I_3 . When the clusters in CG1D evolve back to CG2D, they re-gain the He identities (number of He atoms). The He identity is determined assuming that the coupled evolution between CG1D and CG2D does not alter the shape of He distribution in the cluster subgroup. For a subgroup $CSG(n)$, the shape function $\rho_n(m')$ can be defined as

$$\rho_n(m') = \frac{C(m', n)}{\sum_m C(m, n)}, \quad (29)$$

where m' is the He identity of cluster $He_m.V_n$ in $CSG(n)$, and m denotes all possible He identities in $CSG(n)$. Therefore, if a cluster in CG1D containing $V_c + 1$ vacancies absorbs a SIA, it will go to CG2D, specifically going to the subgroup $CSG(V_c)$. In $CSG(V_c)$, the concentration of clusters containing m' (ranging from 0 to 50 in this study) He atoms will increase by $\rho_n(m') \cdot k_{(V_c+1)}^{SIA} C(V_c + 1) CSIA_s(1)$. In other words, the shape function actually determines the fraction of new clusters with all possible He contents when the clusters evolve back to CG2D. The validation of the coupling scheme is demonstrated in the *Supplementary note 1.2*.

2.5. Effects of impurities on vacancy behavior

Because carbon interacts strongly with vacancies in bcc iron to form thermal stable vacancy-carbon clusters, it can greatly modify the mobility of vacancies. When the material system contains an excess concentration of carbon compared with vacancies, i.e., assuming all vacancies are trapped by carbon, the migration of vacancy requires the dissociation of V-C clusters. For VC_n clusters, VC_2 is shown to be the most stable configuration [44]. The effective vacancy migration energy, then, is the sum of the total binding energy of VC_2 (1.18 eV) and vacancy free migration energy (0.67 eV) [44]. Due to the co-existence of various V-C clusters, which play dominant roles in different temperature ranges, the vacancy diffusivity exhibits non-Arrhenius behavior [44]. Depending on the temperature, the value of effective migration energy of vacancy is expected to vary between 0.67 eV and 1.85 eV.

In Eurofer97, the content of solute carbon atoms in ferritic structures is about 500 ± 200 appm obtained by atom probe tomography (APT) [45]. However, there is no available experimental data about the vacancy migration energy in this carbon content range. Furthermore, it is worth noting that during irradiation, the concentration of solute carbon in Eurofer97 is expected to decrease due to carbon migration to sinks (e.g., grain boundaries). This indicates that the effects of carbon on vacancy diffusivity might even be non-constant during irradiation. As we do not intend to set the vacancy diffusivity as a fitting parameter in the model, we adopted the effective migration energy of vacancy, 1.37 eV, measured in iron containing 100 appm carbon [46]. This experimental value (1.37 eV) was obtained from the iron matrix, the carbon content of which is closest to the carbon content in Eurofer97 among the reported experiments [46–48]. We are aware that a set of experiments also reported values around 1.3–1.4 eV in iron-based steels containing different contents of carbon [47,48].

The alloying elements, such as W, Cr and Mn in Eurofer97 steel, also participate in the interaction of C-V clusters. Experimentally, the thermally stable solute-carbon-vacancy (S-C-V) clusters have been observed using APT [49]. The study by Bakaev et al. using DFT calculations [50] showed that the participation of solute atoms clearly enhanced the binding energies of S-V and C-V clusters. The

energetic stability of larger S-C-V clusters could likely be realized by the further addition of vacancies and carbon atoms. These studies implied that vacancy dissociation from S-C-V clusters would be harder than that from V-C or V-S clusters. Therefore, the nucleation of stable S-C-V clusters, which are responsible for the formation of a large number of carbides found in irradiated Eurofer97, can regard as potential sinks for vacancies. In the model, we add a sink term to the Eq. (20) for vacancies as $S_{S-C-V}^V = 4\pi r_{\text{trap}} C_{S-C-V}$. The concentration of various S-C-V clusters, C_{S-C-V} , and trapping radius r_{trap} are not evaluated here, we take $r_{\text{trap}} C_{S-C-V}$ as a fitting parameter in the model. If setting C_{S-C-V} to be the nominal carbon content, the best fitting of r_{trap} is $1.7a_0$.

2.6. Numerical algorithms

Considering a system to model the evolution of a cluster group with the largest cluster $He_{1000}V_{1000}$ (radius ~ 1.4 nm), more than 1×10^6 ODEs defined in Eq. (6) need to be solved. The computational cost is huge. Here, the model applies an analytical form of Jacobian matrix $J_{ij} = \partial(\frac{dC_i}{dt})/\partial C_j$, where C_i and C_j are the densities of cluster i and j , respectively. This matrix is sparse, as He-V clusters are coupled with each other only through mobile defects of limited types [13]. The numerical solution of Jacobian matrix is calculated using the python wrappers of odespy package [51] based on the odepack algorithms [52]. Though the sparsity of Jacobian matrix, the computation time and memory use still can become too large as the size of evolving clusters increases with irradiation time. Evidently, the task becomes even more challenging when the clustering of self-interstitial clusters/loops with ODEs defined in Eq. (7) is also involved in the system. The application of the coupling scheme can largely ease this challenge, by properly setting the size of CG2D regime.

2.7. Input parameters

The input parameters of the model, including the irradiation condition and material parameters, are shown in Table 1 and Table 2.

3. Results and discussion

3.1. Dynamical He-V clustering and typical defect evolution

Fig. 3 displays the dynamical He-V clustering course in CG2D upon irradiation up to 1.0×10^5 s (cumulative 70 dpa and 1200 appm He). The formation of He-V clusters after irradiation for 1.0×10^2 s concentrates in a small phase space within 5 He atoms and 5 vacancies. The cluster population increases obviously at 1.0×10^3 s, but are confined in a V-shape region due to the cluster intrinsic regulation associated with defect dissociation energies [12,19]. In this period, the clusters are generally in the CG2D regime. With the irradiation going on, to 3.0×10^3 s, the clustering core (the cluster densely populated phase space) expands following a direction, along which the He/V ratio is around 1. From the irradiation time 1.0×10^3 s to 3.0×10^3 s, a part of evolving clusters, with vacancy number over 20, grow out of CG2D and thus enter CG1D. During the further irradiation, from 5.0×10^3 s to 1.0×10^5 s, the overall cluster density in CG2D decreases gradually with the direction of the clustering core shifting towards lower He/V ratios. Though the cluster population decreases, there is still a high density of small clusters in the $He_{n<5}V_{m<5}$ regime. This is caused by the continuous implantation of He atoms, and generation of vacancies and small pure vacancy clusters directly from irradiation intra-cascades clustering, as described by Eq. (12).

Table 1

Irradiation condition. The symbol # in the third column indicates that $f_{V(4)}$ and $f_{SIA(4)}$ include contributions of all larger clusters with components greater than 4.

Symbol	Description	Value	Reference
T	Temperature	673 K	[15]
G_{dpa}	Damage rate	6.94×10^{-4} dpa/s	[15]
G_{He}	He production rate	1.20×10^{-2} appm/s	[15]
η	Damage efficiency	0.2	[31]
$f_{SIA(1)}$	Single interstitial fraction in cascade	0.6	[31]
$f_{SIA(2)}$	Dimer interstitial fraction in cascade	0.13	[31]
$f_{SIA(3)}$	Trimer interstitial fraction in cascade	0.05	[31]
$f_{SIA(4)}$	Four-interstitial fraction in cascade	0.22#	[31]
$f_{V(1)}$	Single vacancy fraction in cascade	0.56	[31]
$f_{V(2)}$	Dimer vacancy fraction in cascade	0.2	[31]
$f_{V(3)}$	Trimer vacancy fraction in cascade	0.06	[31]
$f_{V(4)}$	Four-vacancy fraction in cascade	0.18#	[31]

Table 2

Material parameters. The symbol * indicates that the available experimental values are adopted.

Symbol	Description	Value	Reference
a_0	Lattice parameter	2.842 Å	
E_f^V	Vacancy formation energy	1.53 eV*	[53]
E_f^{SIA}	SIA formation energy	3.77 eV	[21]
E_f^{He}	He formation energy	4.37 eV	[54]
D_0	Pre-exponential factor	$10^{13} a_0^2$	[21]
E_m^V	Vacancy migration energy	1.37 eV*	[46]
E_m^{SIA}	SIA migration energy	0.34 eV	[21]
E_m^{He}	Interstitial He migration energy	0.06 eV	[6]
$E_m^{I_2}$	I_2 migration energy	0.42 eV	[21]
$E_m^{I_3}$	I_3 migration energy	0.43 eV	[21]
$E_{SIA(2)}^{SIA}$	Binding energy of interstitial dimer	0.80 eV	[21]
ρ_{dl}	Dislocation density	$9.0 \times 10^9 / \text{cm}^2$	[14]
b	Burgers vector of the loop assumed to be prismatic	$\sqrt{3}a_0/2$	[14]
θ	Misorientation angle of low-angle GBs	2.5°	[40]
ν	Poisson's ratio	0.3	
μ	Shear modulus	75.38 GPa	
$\Delta\Omega_V$	Relaxation volume of vacancy	-0.05Ω	[55]
$\Delta\Omega_{SIA}$	Relaxation volume of SIA	1.4Ω	[56]
r_c^V	Dislocation core radius for vacancy	8 Å	[57]
r_c^{SIA}	Dislocation core radius for SIA	13 Å	[57]

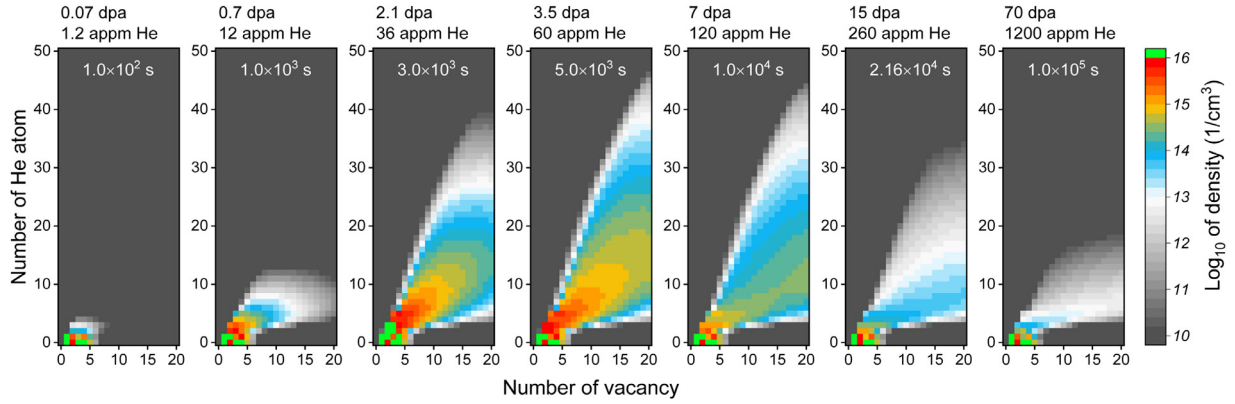


Fig. 3. Sequential snapshots of He-V phase space showing the dynamical He-V clustering. The cluster density scalebar on the right side is valid for all snapshots. He-V clusters with densities over $10^{16}/\text{cm}^3$ are indicated uniformly by the green pixels.

Fig. 4 shows the concentration evolution of typical defects during irradiation. At the very beginning of irradiation, the implanted He atoms occupy mainly interstitial sites due to the low concentration of vacancies. The trapping of interstitial He becomes remarkable as vacancy concentration increases. After a short time (10 s), substitutional He accumulates to a high concentration ($\sim 10^{16}/\text{cm}^3$), which essentially enhances its reaction rate with SIAs. To test the effects of the kick-out reactions on interstitial He, we switch off the reaction ($\text{HeV} + \text{SIA} \rightarrow \text{He}_{\text{int}}$) and re-execute the simulation. The comparison of the concentration profiles obtained in

the simulation with and without the kick-out reactions clearly demonstrates that the operation of kick-out reactions causes the second increase of concentration of interstitial He, which presents two peaks as a function of irradiation time (i.e., irradiation dose). Note that the amount of substitutional He kicked-out by SIAs only accounts for a negligible fraction of its total amount, therefore, the concentration of substitutional He does not show big variations in Phase II. He dissociation from substitutional sites (Frank-Turnbull mechanism) by thermal activation also contributes to the generation of interstitial He atoms, but such an impact is analyzed to be

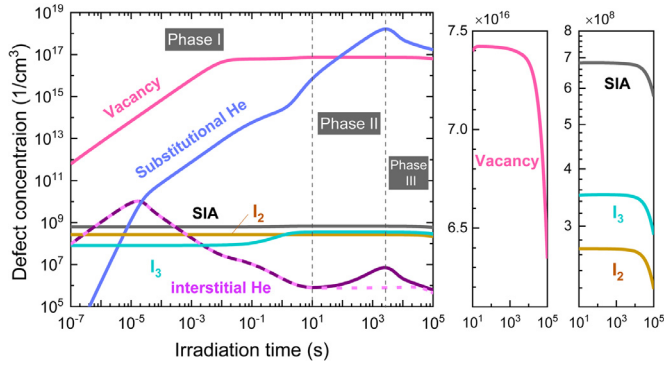


Fig. 4. Concentration evolution of vacancy, SIA, interstitial He, substitutional He, I_2 and I_3 with irradiation time. The dashed line represents the concentration of interstitial He when the kick-out reaction is switched off. Based on the variation of interstitial He concentration, the entire irradiation course is divided into three phases, i.e., Phase I, II and III. Two diagrams on the right show respectively the concentration variation details of vacancy and interstitial-type defects in Phases II and III.

small at 673 K compared with the kick-out mechanism. It is worth noting that the formation and evolution of interstitial-type dislocation loops, as described by Eq. (7), is also considered in the model. Their growth is responsible for the concentration decrease of self-interstitial-type defects (SIA, I_2 and I_3) in the irradiation Phase III.

3.2. Size distribution and growth rate of evolving He-V clusters

Fig. 5(a) shows the evolution of modeled cluster sizes as a function of irradiation time. The classic saddle pattern of size distribution is present after long time irradiation. The size distribution at 2.16×10^4 s (15 dpa and 260 appm He) is shown to compare with available experimental data [15,58]. As shown in Fig. 5(b), the experimental and modeled bubble size distributions at 2.16×10^4 s were plotted using a bin size of 0.05 nm. For small clusters with radii less than 0.4 nm, the experimental data is not available due to the limitation of observation using transmission electron microscopy (TEM). The model predicts the peak radii of He-V clusters being around 0.75 nm, i.e., 1.5 nm in diameter, which is slightly larger than the peak size (1.2 nm) of He bubbles observed experimentally.

Experimentally, a total bubble density of $9.40 \times 10^{16} / \text{cm}^3$ was detected. Considering the sizes of clusters with vacancy number over 20 in CG1D are larger than 0.86 nm, generally resolvable by TEM, the cluster density in CG1D is integrated for comparison, giving a total cluster density of $2.93 \times 10^{17} / \text{cm}^3$. Therefore, the model predicts a two times higher bubble density. This discrepancy

is likely to be caused by (i) the inevitable underestimated number of small bubbles observed by TEM, and (ii) the approximation made in the model, where the loss of He atoms to existing sinks is ignored. Therefore, the model may predict more He-V cluster nuclei, resulting in a bubble density higher than the experimental value. The prediction of the higher bubble size by the model can be attributed to the effective vacancy migration energy (1.37 eV) used in the model. Expectedly, a higher value of migration energy could slow vacancy motion, and thus reduce the bubble growth rate.

To statistically follow the cluster growth rate, a quantity of weighted cluster radius is defined as $r_{wt} = (\sum r_i^3 C_i / \sum C_i)^{1/3}$, where r_i and C_i are the cluster radius and corresponding density, respectively. As shown in Fig. 6(a), the weighted radii of clusters in CG1D and CG2D+CG1D increase monotonically in the irradiation Phases II and III (from 10 s to 10^5 s). The weighted cluster radius of CG2D+CG1D (all He-V clusters) is smaller than that of CG1D until the irradiation time 1.2×10^4 s, because initially the small clusters dominate the whole cluster population. Afterwards, two curves are overlapped since the cluster density in CG2D gradually decreases due to the cluster growth induced cluster transfer from CG2D to CG1D. The weighted cluster radius in CG1D obtained in the simulation without the kick-out reaction is also shown for comparison. One can find that cluster sizes are smaller without the production of the kicked-out He during irradiation.

By laying tangent to the curve, the cluster growth rate V_r is obtained in the unit of nm/s (Fig. 6(b)). Surprisingly, two rate peaks appear in the simulation considering the kick-out reaction. The inset figure illustrates that the first peak is associated with the concentration increase of interstitial He. This indicates that, because of kick-out reactions, more He atoms can flow to clusters and thus enhance the growth of clusters. On the one hand, absorption of He can directly increase cluster size (Fig. S1(c)), though the impact is small compared with the absorption of vacancy. On the other hand, the absorption of He would increase cluster He/V ratio, which reduces the emission of vacancy and increase the emission of SIA (Fig. S1(d)). These two aspects both contribute to the increase of V_r before 1.2×10^4 s. The latter aspect plays a major role in controlling the trend of the overall increase of V_r , while the first aspect causes the small bump in V_r . After 1.2×10^4 s, several factors result in the decrease of V_r : 1) decrease of cluster He/V ratio (Fig. 7), 2) high number density of clusters in CG1D (Fig. 6), and 3) the decrease of He and vacancy concentration (Fig. 4). It is worth noting that the coupling between CG2D and CG1D also affect V_r in CG1D, since clusters continuously entering CG1D would change the weighted radius r_{wt} especially before 1.2×10^4 s. The simulation without the kick-out reaction, however, shows rather stable

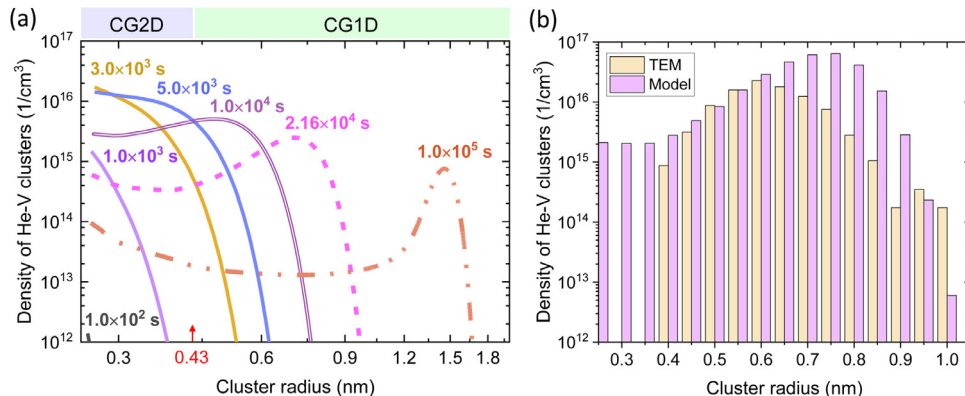


Fig. 5. Size distribution of He-V clusters. (a) Evolution of cluster size distribution with irradiation time corresponding to those in Fig. 3. The boundary of two cluster groups (CG2D and CG1D) is located at 0.43 nm. (b) Comparison of cluster size distribution obtained by TEM observation [15,58] and modeling at 2.16×10^4 s.

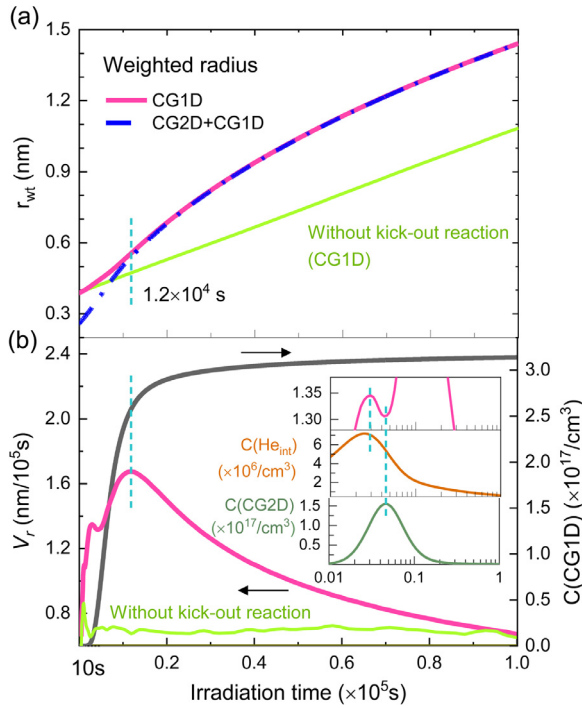


Fig. 6. Weighted radius and growth rate of clusters as a function of time. (a) Weighted cluster radius (r_{wt}) in CG1D and CG2D+CG1D. For the simulation without the kick-out reaction, r_{wt} is estimated for the clusters in CG1D only. (b) Growth rate (V_r) and number density of clusters in CG1D. V_r obtained in the simulation without the kick-out reaction is shown for comparison. The inset displays the magnified locations of the initial two rate extrema, and compared with the concentration curve of interstitial He (He_{int}) and density curve of He-V clusters (excluding $\text{He}_{n<5}\text{V}_{m<5}$) in CG2D.

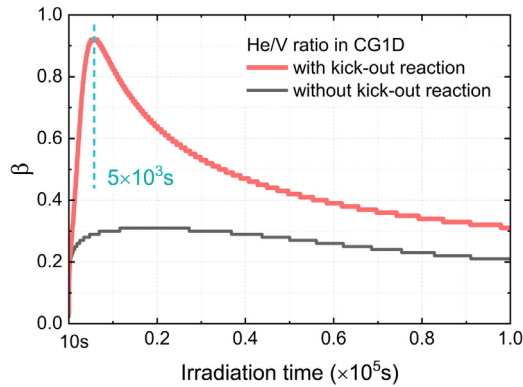


Fig. 7. Dynamical cluster He/V ratio (β) in CG1D as a function of irradiation time. β obtained in the simulation without the kick-out reaction is also shown for comparison.

V_r in the period of cluster coupled evolution. This suggests the insignificant effect of the coupling scheme on V_r . Without the kick-out reaction, V_r is much lower, supporting the conclusion that the kick-out reaction promotes the growth of He-V clusters. The stable V_r can be attributed to the nearly constant cluster He/V ratio (with variation in the range of ~ 0.2 - 0.3) during irradiation (Fig. 7). Note that the calculated growth rate reflects the nominal growth behavior of a cluster group instead of individual clusters. The growth rate is expected to vary with different cluster group sizes.

3.3. He/V ratio and pressurization state of evolving He-V clusters

Fig. 7 shows the evolution of cluster He/V ratio in CG1D as a function of irradiation time obtained in simulations with and with-

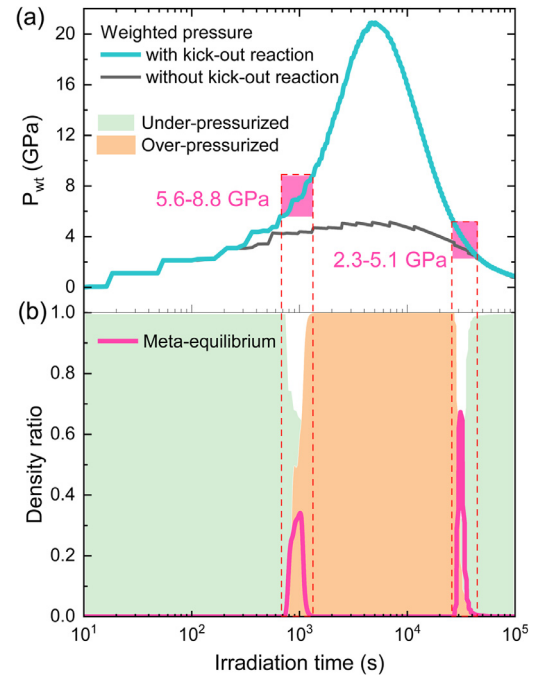


Fig. 8. Pressurization state of evolving clusters in CG1D. (a) Weighted cluster pressure (P_{wt}). P_{wt} obtained in the simulation without the kick-out reaction is shown for comparison. (b) Density ratio of He-V clusters in each pressurization state (under-pressurized, over-pressurized and meta-equilibrium).

out the kick-out reaction. With the kick-out reaction, the cluster He/V ratio in CG1D rises rapidly from the onset of irradiation to $\sim 5 \times 10^3$ s. The increase happens in a period covering the time range of irradiation Phase II ($0 - 2.2 \times 10^3$ s). In Phase II, the vacancy concentration is high but relatively constant as a function of time (Fig. 4), while the concentration of interstitial He varies in the range from 10^6 to 10^7 /cm³ because of the kick-out mechanism. The concentration increase is about 1 order of magnitude, indicating that the absorption rate of He atoms by clusters would be enhanced by an equivalent amount. Without the kick-out reaction, the cluster He/V ratio in CG1D is rather stable varying in the range between ~ 0.2 and 0.3 . This suggests the kick-out mechanism also plays a major role in the evolution of cluster He/vacancy ratios. It is worth noting that the evolving range of He/V ratio along with the weighted cluster radius, consists well with several experimental measurements for He bubbles in ferritic/martensitic steels [59–61]. For example, He density in a 1.3 nm bubble was measured in Ref. [59], leading to a He/V ratio ~ 0.6 , and in Ref. [60] the He/V ratio for larger He bubbles between 2 and 5 nm was estimated in the range of 0.25–0.85.

To explore the cluster pressurization states, also in a statistical way, the weighted cluster pressure is calculated as $P_{wt} = \sum P_i C_i / \sum C_i$, where P_i is the pressure of individual clusters. The cluster pressure is calculated based on a four-term virial expansion of the compressibility factor by Trinkaus [28] (also see *supplementary note 1.1*). Fig. 8(a) shows that the variation of P_{wt} can exceed 20 GPa when the kick-out reaction is considered, while the clusters evolve in the pressure range below 6 GPa in the simulation without the kick-out reaction. The large pressure variation of clusters is expected to affect significantly their pressurization states.

Applying the Young-Laplace equation [62], the thermal equilibrium pressure of clusters can be estimated using $P_{eq} = 2\gamma/r$ with $\gamma = \gamma_\infty / (1 + \delta/r)$, where γ is free energy of the He-metal interface, r is cluster radius. γ_∞ is the surface energy for a flat surface, 1.98 J/m² in bcc iron adopted in VGM [18]. δ is the Tolman length quantified essentially as the gap size between He atoms and the ma-

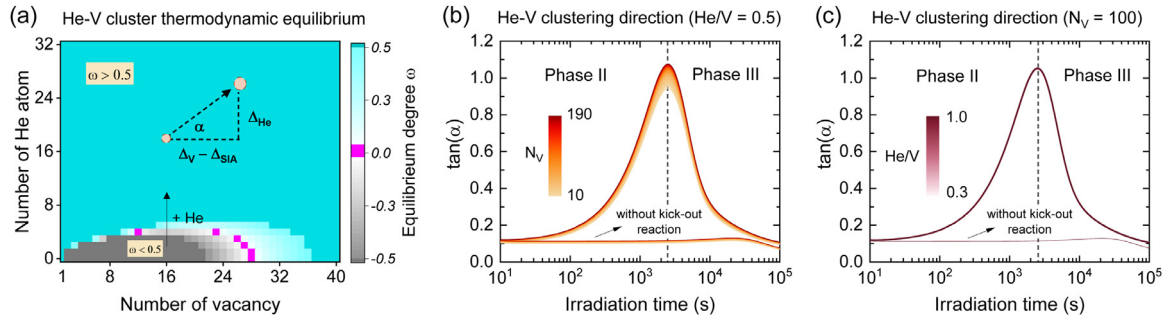


Fig. 9. Interstitial He governs the He-V clustering direction. (a) Thermodynamic equilibrium degree ω of He-V clusters at 2.2×10^3 s. He-V clustering direction is quantified as $\tan \alpha$ for (b) clusters of a given He/V ratio 0.5 with the vacancy number N_V ranging from 10 to 190, and (c) for clusters of a given vacancy number, $N_V = 100$, with cluster He/V ratio ranging from 0.3 to 1.0. The $\tan(\alpha)$ curves for the clusters with different He/V ratios (0.3–1.0) are strongly overlapped in (c). $\tan(\alpha)$ obtained in the simulation without the kick-out reaction is also shown in (b) and (c) for comparison.

trix in VGM. Additionally, we define a meta-equilibrium state, in which the cluster pressure is within $\pm 5\%$ deviation range of the corresponding P_{eq} . Beyond this range, He-V clusters are assumed to be either in the over-pressurized state (Over the upper limit (+5%)) or under-pressurized state (below the lower limit (−5%)). Under this definition, the results show that He-V clusters are initially under-pressurized (Fig. 8(b)). Due to the fast-increasing pressures caused by the enhanced absorption of kicked-out interstitial He during the irradiation Phase II, He-V clusters shift swiftly into the over-pressurized state. Finally, the clusters again return to the under-pressurized state as the pressure decreases. With the growing of clusters, the cluster pressures (2.3–5.1 GPa) at the second state transition were lower than those (5.6–8.8 GPa) at the first one (Fig. 8(a)).

3.4. Thermodynamic equilibrium and clustering direction

The Young-Laplace equation, using the surface energy, expresses the equilibrium of vacancies between the clusters and matrix. Such an equilibrium state, depending only on the cluster size and matrix surface energy, shows no links to the external irradiation conditions. To quantify the cluster thermodynamic equilibrium degree during the irradiation, we define a quantity ω as $\omega = (\Delta_V / \Delta_{SIA}) - 1$, based on the ability of the cluster absorbing vacancies (Δ_V) and SIAs (Δ_{SIA}). For a He-V cluster He_mV_n , neglecting the mobilities of I_2 and I_3 for simplicity, Δ_V and Δ_{SIA} can be estimated as $\Delta_V = k_{(m,n)}^V \times C_{eq}^V - g_{(m,n)}^V$ and $\Delta_{SIA} = k_{(m,n)}^{SIA} \times C_{eq}^{SIA} - g_{(m,n)}^{SIA}$, respectively. The value $\omega = 0$ indicates that the cluster is at the thermodynamic equilibrium state (TES). With a positive ω , the cluster will increase its vacancy content (grow large), either by the emission of SIA or/and capture of vacancy depending on the cluster internal configuration (He/V ratio) and external concentrations of SIA and vacancy in the matrix. Vice versa for the cluster with negative ω . The similar concept was applied previously by Russel to describe the nucleation trajectories of gas-filled cavities [63]. The velocities of the cavity capturing vacancies and gas atoms were separately quantified based on the rates of defects reaching and emitting from the cavity. Here, the role of He atoms was not involved in the estimation of ω , it will be discussed later in terms of another defined quantity (clustering direction).

If a given He-V cluster is deliberately inserted in a thermal equilibrium matrix at 673 K (See *supplementary note 1.3*), the phase space for clusters staying at $|\omega| \leq 0.5$ displays a channel shape (Fig. S4). Under irradiation, for estimating ω the equilibrium concentration of defects C_{eq}^x was replaced by the real concentration in the matrix depending on irradiation dose (time). For example at the irradiation time of 2.2×10^3 s, applying the above calculation scheme, results show that the equilibrium channel becomes a closed “equilibrium hill” (Fig. 9(a)). When a cluster captures mo-

bile He atoms, as indicated by the arrow in Fig. 9(a), it evolves out of the “hill” and will always hold positive ω .

He-V clusters away from the TES gain the driving force for evolving. The evolving (or clustering) direction, as schematically shown in Fig. 9(a), can be quantified as $\tan \alpha = \Delta_{He} / (\Delta_V - \Delta_{SIA})$. Δ_{He} can be estimated in a same manner as for Δ_V and Δ_{SIA} . Using the concentration data of interstitial He, vacancy and SIA in Fig. 4, $\tan \alpha$ is estimated for a set of He-V clusters, as shown in Fig. 9(b) and 9(c). As a comparison, $\tan \alpha$ is also calculated for the simulation without the kick-out reaction to show the role of the kick-out reaction in mediating the clustering direction. Without the kick-out reaction, the calculated $\tan \alpha$ is rather stable (with the value of ~ 0.1) in the irradiation Phases II and III. On the contrary, $\tan \alpha$ obtained in the simulation considering the kick-out reaction shows big variations. Moreover, it is surprised to see that the variation of $\tan \alpha$ displays a similar trend with the concentration of interstitial He in the irradiation Phases II and III. This implies that the operation of the kick-out mechanism also plays a decisive role in determining the clustering directions with the participation of vacancies and SIAs. Furthermore, it appears that the clustering directions exhibit a rather weak dependence on cluster sizes (Fig. 9(b)). More interestingly, for a set of clusters containing 100 vacancies, their clustering directions are insensitive to the cluster He/V ratio in the range of 0.3–1.0 (Fig. 9(c)). Since the He/V ratio regulates the defect emission rate of the clusters, such a weak dependency implies that the defect emission (outflow) from He-V clusters plays a minor role in the cluster evolution. In contrast, the strong defect absorption (inflow) by (to) He-V clusters dominates the cluster evolution course.

4. Summary of the main assumptions in the model

4.1. Homogeneous nucleation of He-V clusters in the matrix

The assumption on homogeneous nucleation of He-V clusters was made based on the experimental observation reported in Refs. [15,58]. He bubbles formed at irradiation temperatures of 603 K, 673 K and 773 K were characterized using TEM by Kaiser [58]. The results indicated a spatially homogeneous distribution of He bubbles at 603 K and 673 K, whereas a coexistence of homogeneous and heterogeneous nucleation of bubbles was found at 773 K. In this work, we focused on the irradiation case at 673 K only.

It can be expected that bubble nucleation at sinks, such as dislocations, grain boundaries and impurities can occur even at lower irradiation temperatures. However, compared with the homogeneous nucleation in the matrix, the heterogeneous bubble evolution is relatively weak at low temperatures.

4.2. Dissociation of defect cluster from large clusters

The assumption that only point defects can be emitted from defect clusters is a widely accepted treatment for simplicity in cluster dynamics modeling [3,8,10–13,20]. The reason can be inferred from the binding energy between defect clusters, as it determines the emission rate of a defect (cluster) from its “mother” cluster.

For self-interstitial clusters/loops, the emission of single point defects (vacancy and SIA) is energetically favorable compared with the emission of defect clusters. This is partly due to the higher formation energy of defect clusters than single point defects in bcc iron. Taking the emission of a I_2 from a $1/2\langle 111 \rangle$ loop containing 10 SIAs as an example, the binding energy can be calculated using the equation, $E_{SIAs(10)}^{I_2} = E_f^{I_2} + E_f^{SIAs(8)} - E_f^{SIAs(10)}$, with formation energy of $E_f^{I_2} = 8.63$ eV, $E_f^{SIAs(8)} = 24.22$ eV and $E_f^{SIAs(10)} = 28.29$ eV [64]. The calculated binding energy $E_{SIAs(10)}^{I_2}$ is 4.56 eV, much higher than $E_{SIAs(10)}^{SIA}$, which is calculated to be 2.09 eV (2.18 eV when using the $n^{2/3}$ extrapolation law (Eq. (14)). For tri-interstitials and larger clusters, expected higher binding energy to clusters/loops indicates weaker emissions.

For He-V clusters, the emission of defect clusters (usually self-interstitials) is likely to happen when a bubble is in the overpressurized state. This kinetic process was called loop punching, a special mechanism for bubble growth [28,65]. Trinkaus et al. studied the energetics involved in loop punching by He bubbles [28]. For small bubble sizes, it was analyzed that the process of loop punching starts with the formation of single interstitials and is followed by their agglomeration into small three-dimensional clusters staying near bubble surface. When the number of interstitials is enough to form an equatorial loop, it is energetically more favorable to transform the three-dimensional cluster into a loop which will be expelled if a critical pressure in the bubble is reached. However, for large bubbles, it is not clear whether this pre-stage of a three-dimensional interstitial cluster is required for loop formation. The details of the loop formation close to bubble surface are not yet fully understood. It is not clear whether, for example, the energy barrier for bubble growth by loop punching coincides with forming a complete dislocation loop around the bubble circumference or with detaching the loop platelet from the bubble surface [65]. The lacking information makes it difficult to implement such a kinetic process in a cluster dynamics model. Furthermore, if considering the initial stage of forming a prismatic dislocation loop near the bubble surface, the distinction between self-interstitial emission (considered in the model) and loop nucleation disappears. More details about the loop punching mechanism can be found in Ref. [65].

4.3. Mobility of large defect clusters

It is theoretically [66] and experimentally [67] proven that interstitial loops are mobile in high-purity iron. However, the presence of impurities and He atoms could significantly affect loop mobility [22–24]. A recent *in situ* annealing work by Yuan et al. shown that dislocation loops in Eurofer97 steels are rather stable up to 550°C which was partly attributed to decoration of the loops by existing impurities [68]. An earlier work by Moll et al. also observed immobility of dislocation loops in He ion-irradiated iron during *in situ* annealings. The reason was attributed to the presence of impurities and He atoms. The assumption of loop trapping by He atoms has been made in earlier cluster dynamics simulations [24]. For all these reasons, no loop mobility is considered in the present calculations.

Large vacancy clusters and He-V clusters are assumed to be immobile based on the fact that 1) diffusivity of cluster decreases

with increasing cluster sizes, and 2) impurities generally play a role in enhancing migration energy of clusters. The impurities here mainly are Cr and carbon atoms. It was reported that the addition of Cr solutes in iron can effectively suppress He desorption during post-irradiation annealing, probably due to the trapping of He-V clusters by Cr solutes [26]. For effects of carbon, as addressed while discussing the migration of vacancy, carbon interacts strongly with vacancies in bcc iron and thus form thermal stable vacancy-carbon clusters, leading to the enhanced migration energy of vacancies [44]. A similar effect of carbon on the migration of He-V clusters can be expected, since Fu et al. reported the formation of helium-carbon-vacancy clusters [25]. The trapping effects would be stronger with the increase of cluster sizes. Therefore, mobility of pure vacancy clusters and He-V clusters was neglected in the model.

4.4. Interaction between He atoms and interstitial dislocation loops

This assumption is also related to the issue of the homogeneous nucleation of bubbles in the matrix. Bubble nucleation at dislocation loops has been reported by many studies, for instance in Refs. [69,70]. As observed by Brimal et al. [70], the high rate of He implantation at 773 K caused the nucleation of He bubbles inside dislocation loops in bcc iron. In Fe-Cr ferritic steels after He ion implantation, Chen et al. observed the formation of bubble-loop complexes [69]. It was found that He bubbles can promote loop growth possibly by an exchange of atoms in the bubble-loop complexes from the bubble surface to the dislocation cores [71,72]. The corresponding interaction kinetics were not considered in the current work, since the interaction between He atoms and dislocation loops is not fully clear in the iron-based matrix to allow accurate determination of the capture (emission) efficiency of He atoms by (from) dislocation loops. Moreover, the growth of dislocation loops and the bubbles attached is expected to have a competition for external vacancies during irradiation. To our knowledge, this kinetics is not fully understood yet.

In our model, we made a simple assumption that bubble evolution away from and near (at) loops is similar, i.e., homogeneous bubble evolution in the matrix. This assumption was made, as aforementioned, based on the TEM observation that bubble evolution at 673 K is likely to be dominated by the homogeneous bubble growth in the matrix [15,58]. Furthermore, no correlation between the spatial distribution of He bubbles and dislocation loops has been reported in Refs. [15,58].

5. Conclusion

He-V clustering in ferritic/martensitic Eurofer97 steel under He^+/Fe^{3+} dual-beam irradiation was investigated by a cluster dynamics model based on rate theory. The modeled cluster size, density and He/V ratio show good consistency with the available experimental values. A comprehensive picture of He-V clustering during continuous He and dpa generation is given, including the evolution of cluster size distribution, cluster growth rate, cluster pressurization state and clustering direction. This work demonstrates that the operation of the kick-out reaction can significantly raise the concentration of interstitial He, leading to the appearance of a second concentration increase (peak) during irradiation. It is unveiled that the kicked-out He atoms significantly promote the growth of He-V clusters, and cause the increase in cluster pressures. The analysis on He-V clustering direction indicates that the kick-out reaction also plays a dominant role in controlling the evolution of cluster configuration (He/V ratio), while the cluster original configuration and size play a minor role.

Credit Author Statement

J.G. and B.K. wrote the codes. E.G. supervised the entire work. J.G performed the calculation and data analysis, and drafted the manuscript. B.K., E.G. and J.A. revised the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Supplementary materials

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